

Assignment Four for Machine Learning Using Python Programming

1. What does the ROC curve plot? a) Precision vs. Recall
b) True Positive Rate vs. False Positive Rate
c) Sensitivity vs. Specificity
d) Accuracy vs. Error

Answer: b) True Positive Rate vs. False Positive Rate

Explanation: The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at different classification thresholds. It helps in evaluating a classifier's performance.

2. What is the significance of AUC (Area Under the Curve)?
a) It represents the proportion of correctly classified instances
b) It measures the overall performance of a classification model
c) It indicates how many false positives exist in the model
d) It is a measure of bias in the dataset

Answer: b) It measures the overall performance of a classification model

Explanation: AUC quantifies the overall ability of a model to distinguish between classes. A higher AUC means better performance. AUC = 1 represents a perfect model, while AUC = 0.5 indicates random guessing.

3. Which classification threshold leads to more false positives?
a) High threshold
b) Low threshold
c) Threshold does not impact false positives
d) Depends on the model

Answer: b) Low threshold

Explanation: Lowering the classification threshold increases the number of positive predictions, which can lead to more false positives.

4. A model with an AUC value of 0.5 indicates:
a) A perfect classifier
b) A completely random model
c) A poor classifier
d) A classifier with 100% accuracy

Answer: b) A completely random model

Explanation: AUC = 0.5 means the model performs no better than random chance. AUC closer to 1 indicates a good classifier, while AUC closer to 0 suggests a poor classifier.

5. In simple linear regression, the dependent variable is also known as:
a) Predictor
b) Independent variable
c) Response variable
d) Feature

Answer: c) Response variable

Explanation: The dependent variable (Y) is the one that we are trying to predict or explain in regression analysis.

MCQs from Lecture 10: Regression Models

6. What is the equation of a simple linear regression model?

- a) $y = mx + b$
- b) $y = b_0 + b_1x$
- c) $y = ax^2 + bx + c$
- d) $y = wx + b + c$

Answer: b) $y = b_0 + b_1x$

Explanation: Simple linear regression follows the equation $y = b_0 + b_1x$, where b_0 is the intercept and b_1 is the slope (coefficient of x).

7. What does the coefficient (b_1) in simple linear regression represent?

- a) Intercept of the line
- b) Rate of change of the dependent variable with respect to the independent variable
- c) The total variance in the dataset
- d) The error term

Answer: b) Rate of change of the dependent variable with respect to the independent variable

Explanation: The coefficient b_1 represents how much Y (dependent variable) changes for every unit increase in X (independent variable).

8. Which of the following is NOT an assumption of linear regression?

- a) Linearity
- b) Homoscedasticity
- c) Multicollinearity
- d) Decision boundary

Answer: d) Decision boundary

Explanation: Decision boundary is a concept from classification models, not regression. Linear regression assumes linearity, normality of residuals, homoscedasticity, and independence of errors.

9. What is the purpose of the least squares method in linear regression?

- a) To maximize errors
- b) To minimize the sum of squared residuals
- c) To find the median of residuals
- d) To maximize variance

Answer: b) To minimize the sum of squared residuals

Explanation: The least squares method minimizes the squared differences between observed values and predicted values, ensuring the best-fitting line.

MCQs from Lecture 11: Gradient Descent

10. What is the goal of gradient descent?

- a) To find the highest point of a function
- b) To find the minimum value of a cost function
- c) To increase the complexity of the model

d) To randomly update model parameters

Answer: b) To find the minimum value of a cost function

Explanation: Gradient descent is an optimization algorithm used to minimize the cost function by updating model parameters iteratively.

11. What is the significance of the learning rate (α) in gradient descent?

a) It controls the step size for parameter updates

b) It determines the number of epochs

c) It controls the dataset size

d) It is irrelevant in optimization

Answer: a) It controls the step size for parameter updates

Explanation: The learning rate (α) determines how much the model updates parameters in each iteration. A high α may lead to overshooting, while a low α leads to slow convergence.

12. Which type of gradient descent uses the entire dataset to compute the gradient?

a) Stochastic Gradient Descent (SGD)

b) Mini-Batch Gradient Descent

c) Batch Gradient Descent

d) None of the above

Answer: c) Batch Gradient Descent

Explanation: Batch Gradient Descent computes the gradient using the entire dataset, making it computationally expensive for large datasets.

13. What is the main disadvantage of Stochastic Gradient Descent (SGD)?

a) It requires large datasets

b) It is computationally expensive

c) It has high variance and noisy updates

d) It does not converge

Answer: c) It has high variance and noisy updates

Explanation: SGD updates parameters after each training example, making it computationally efficient but also prone to fluctuations and noise.

14. Which method helps accelerate gradient descent convergence?

a) Increasing the learning rate

b) Adding momentum

c) Reducing batch size

d) Ignoring cost function

Answer: b) Adding momentum

Explanation: Momentum helps smooth updates by incorporating previous gradients, reducing oscillations, and speeding up convergence.

15. What happens if the learning rate in gradient descent is too high?

a) The model converges too quickly

b) The model may overshoot and fail to converge

c) The model will always find the global minimum

d) The model will have zero training error

Answer: b) The model may overshoot and fail to converge

Explanation: A very high learning rate can cause the algorithm to take large steps, overshooting the minimum of the cost function and leading to divergence rather than convergence. A balanced learning rate ensures stable and efficient convergence.

16. Which of the following statements is false about gradient descent?
- a) It updates the weight to comprise a small step in the direction of the negative gradient
 - b) The learning rate parameter is η where $\eta > 0$
 - c) In each iteration, the gradient is re-evaluated for the new weight vector
 - d) In each iteration, the weight is updated in the direction of positive gradient

Ans: d) In each iteration, the weight is updated in the direction of positive gradient

Explanation: Gradient descent is an optimization algorithm, and in each iteration the weight is not updated in the direction of positive gradient. Here it updates the weight in the direction of the negative gradient. And the gradient is re-evaluated for the new weight vector with a learning parameter $\eta > 0$.

17. What is the gradient of the function $2x^2 - 3y^2 + 4y - 10$ at point $(0, 0)$?
- a) $0i + 4j$
 - b) $1i + 10j$
 - c) $2i - 3j$
 - d) $-3i + 4j$

Ans: a) $0i + 4j$

Explanation: Given the function $f = 2x^2 - 3y^2 + 4y - 10$ at point $(0, 0)$. Then the gradient of the function can be calculated as:

$$\begin{aligned}\partial f / \partial x &= \partial(2x^2 - 3y^2 + 4y - 10) / \partial x \\ &= 4x \\ &= 4 * 0 \\ &= 0 \\ \partial f / \partial y &= \partial(2x^2 - 3y^2 + 4y - 10) / \partial y \\ &= -6y + 4 \\ &= (-6 * 0) + 4 \\ &= 0 + 4 \\ &= 4\end{aligned}$$

Gradient, $\nabla f = 0i + 4j$

18. In Scikit-learn, which class is used for implementing Linear Regression?
- a) `sklearn.linear_model.LinearRegression`
 - b) `sklearn.regression.LinearModel`
 - c) `sklearn.model_selection.LinearRegression`
 - d) `sklearn.preprocessing.LinearRegressor`

Ans: `sklearn.linear_model.LinearRegression`

Explanation: In scikit-learn, `sklearn.linear_model.LinearRegression` is a class that implements linear regression.

19. Which evaluation metric is commonly used to measure the performance of a Linear Regression model?
- a) Mean Absolute Error (MAE)

- b) Mean Squared Error (MSE)
- c) R-squared (R^2)
- d) All of the above

Ans: d) All of the above

Explanation: The most commonly used evaluation metric for a Linear Regression model is the Mean Squared Error (MSE), which measures the average of the squared differences between the actual values and the predicted values by the model; often, the square root of MSE, known as Root Mean Squared Error (RMSE), is also used for interpretation purposes.

20. What does a high R-squared (R^2) value indicate in Linear Regression?

- a) The model is underfitting
- b) The model explains most of the variance in the data
- c) The model is overfitting
- d) The model is not useful

Ans: b) The model explains most of the variance in the data

Explanation: A high R-squared value in linear regression indicates that a large proportion of the variance in the dependent variable is explained by the independent variables in the model